



2009
HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION

Extension 2 Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question
- Start a new booklet for each question

Total marks – 120

- Attempt Questions 1 – 8
- All questions are of equal value

Total Marks – 120

Attempt Questions 1-8

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1.	(15 marks)	Start a new page	Marks
(a) (i)	Show that $y = x\sqrt{4-x^2}$ is an odd function.		1
(ii)	Hence without finding the integral evaluate $\int_{-2}^2 \left(x\sqrt{4-x^2} - \sqrt{4-x^2} \right)$, giving reasons.		2
(b)	By using the table of standard integrals, find $\int \frac{dx}{\sqrt{4x^2+36}}$		2
(c)	Use partial fractions to evaluate $\int_0^1 \frac{5dt}{(2t+1)(2-t)}$		3
(d)	Find $\int \operatorname{cosec} x \, dx$ by using the substitution $t = \tan \frac{x}{2}$		3
(e)	Find $\int \frac{\sqrt{x^2-16}}{x} \, dx$ using the substitution $x = 4\sec\theta$.		4

End of Question 1

Question 2. (15 marks) Start a new page

Marks

(a) Express $\frac{2-5i}{4-3i}$ in the form $x + iy$ where x and y are real. 2

(b) Find all pairs of integers for a and b such that $(a - ib)^2 = -21 - 20i$ 3

(c) Find the modulus and argument of $(\sin \theta + i \cos \theta)(\cos \theta - i \sin \theta)$ 3

(d) (i) If $\left| \frac{z-1}{z+1} \right| = 2$, where $z = x + iy$, show that the locus of z is 2

$$\left(x + \frac{5}{3} \right)^2 + y^2 = \frac{16}{9}$$

(ii) Represent this locus on an Argand Diagram and shade the region 3
for which the inequalities $\left| \frac{z-1}{z+1} \right| \leq 2$ and $0 \leq \arg z \leq \frac{3\pi}{4}$ are both
satisfied.

(e) z_1 and z_2 are two complex numbers such that $\frac{z_1 + z_2}{z_1 - z_2} = 2i$ 2

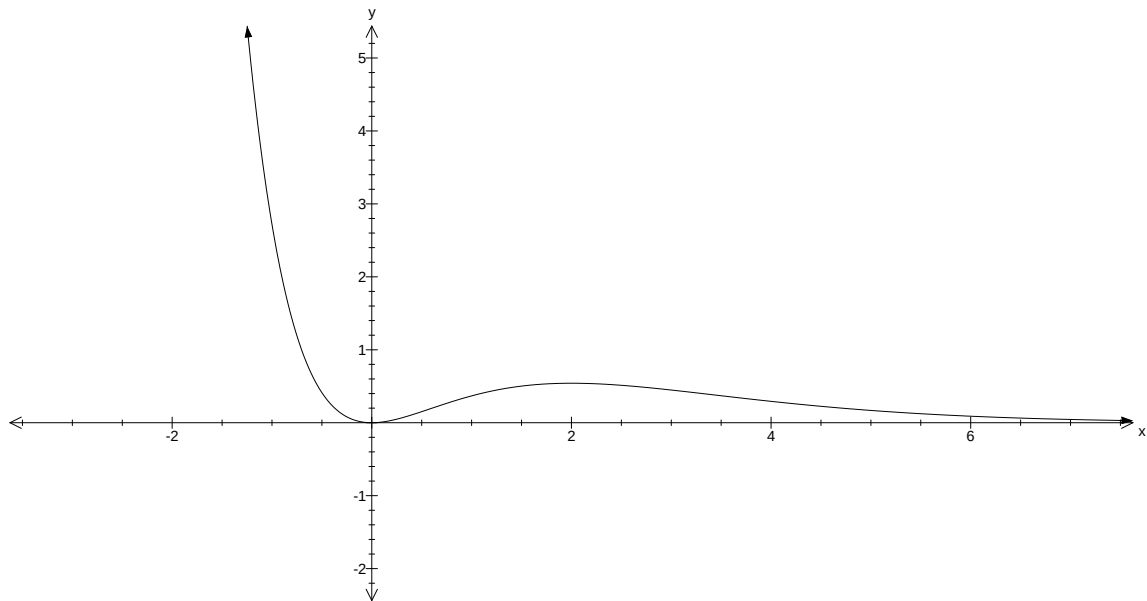
On an Argand diagram show vectors representing $z_1, z_2, z_1 + z_2$ and $z_1 - z_2$

End of Question 2

Question 3. (15 marks) [Start a new page](#)

Marks

(a)



The graph of $y = x^2 e^{-x}$ is sketched above. There is a stationary point at $(0,0)$ and $\left(2, \frac{4}{e^2}\right)$

On separate diagrams, draw a neat sketch showing the main features of each of the following

(i) $y = f(x) + 1$ 1

(ii) $y = f(|x|)$ 1

(iii) $y = \{f(x)\}^2$ 2

(iv) $y = \frac{1}{f(x)}$ 2

(v) $y^2 = f(x)$ 2

(vi) $y = \cos^{-1}(f(x))$ 2

(b) If $x^m y^n = k$, where k is a constant, show that $\frac{dy}{dx} = -\frac{my}{nx}$ 2

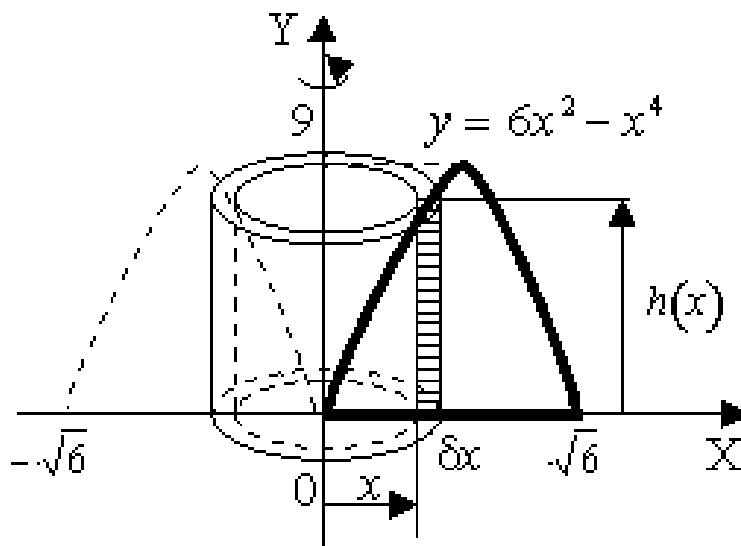
Question 3 continues on page 5

Question 3 continued

Marks

- (c) Using the method of cylindrical shells find the volume of the solid of revolution generated when the area enclosed by the curve $y = 6x^2 - x^4$ the x -axis and $0 \leq x \leq \sqrt{6}$ is rotated about the y - axis.

3



End of Question 3

Question 4. (15 marks) Start a new page

Marks

(a) If α, β, γ are the roots of the equation $x^3 - 4x^2 + 2x + 5 = 0$. Evaluate:

(i) $\alpha^2 + \beta^2 + \gamma^2$ **1**

(ii) $\alpha^3 + \beta^3 + \gamma^3$ **2**

(b) $P(x)$ is a monic polynomial of degree 4 with integer coefficients and constant term 4. **3**
One zero is $\sqrt{2}$, another zero is rational and the sum of the zeros is positive.
Factorise $P(x)$ fully over $\mathbf{R}$.

(c) (i) Use De Moivre's theorem to show $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$ **3**

(ii) Hence solve $8x^3 - 6x - 1 = 0$ leaving answer in terms of $\cos \theta$ **3**

(d) For a real number r , the polynomial $8x^3 - 4x^2 - 42x + 45$ is divisible by $(x - r)^2$. **3**
Find the value of r .

End of Question 4

Question 5. (15 marks) Start a new page

Marks

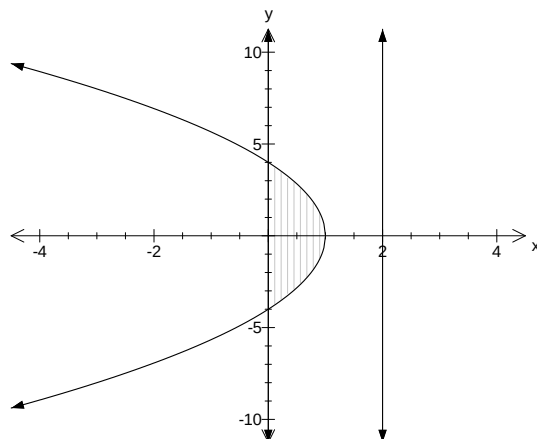
- (a) Evaluate $\int_1^{\infty} \frac{1}{x+1} - \frac{1}{x+3} dx$ 2
- (b) (i) Use integration by parts to show that a reduction (recurrence) formula 3
for $I_n = \int \sin^n x dx$ is $I_n = -\frac{1}{n} \cos x \sin^{n-1} x + \frac{n-1}{n} I_{n-2}$
- (ii) Hence evaluate $\int_0^{\frac{\pi}{2}} \sin^4 x dx$ 2
- (c) The hyperbola H has equation $xy = 4$.
- (i) Sketch the hyperbola and indicate on your diagram the position and coordinates of all points at which H intersects the axes of symmetry. 1
- (ii) Show that the equation of the tangent at $P\left(2t, \frac{2}{t}\right)$ where $t \neq 0$, is $x + t^2 y = 4t$ 2
- (iii) If $s \neq 0$ and $s^2 \neq t^2$, show that the tangents to H at P and $Q\left(2s, \frac{2}{s}\right)$ intersect at 2
 $M\left(\frac{4st}{s+t}, \frac{4}{s+t}\right)$
- (iv) Suppose that in (iii) the parameter $s = -\frac{1}{t}$. Show that the locus of M is a straight 3
line through, but excluding the origin.

End of Question 5

Question 6. (15 marks) Start a new page

Marks

(a)



A solid S is formed by rotating the region bounded by the parabola $y^2 = 16(1-x)$ and the y -axis around the line $x = 2$.
By using the method of slices find the exact volume of S .

4

(b) A hyperbola has foci $(\pm 10, 0)$ and asymptotes $y = \pm \frac{4x}{3}$.

(i) Find the eccentricity.

1

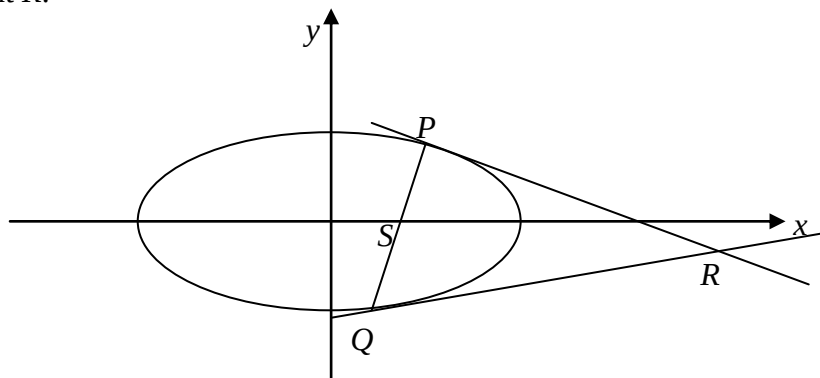
(ii) State the equation of the hyperbola.

1

(iii) Sketch the hyperbola indicating important features such as vertices, foci, directrices and asymptotes

2

(c) Let $P(a \cos \theta, b \sin \theta)$ and $Q(a \cos \phi, b \sin \phi)$ be points on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the extremities of a focal chord PQ . The tangents drawn from the extremities intersect at a point R .



(i) Show that the tangent at P is given by $\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1$.

2

Question 6 continues on page 9

Question 6 continued

Marks

- (ii) Use simultaneous equations to show that the x coordinate of the point R is given **2**

$$\text{by } x = \frac{a(\sin \phi - \sin \theta)}{\cos \theta \sin \phi - \sin \theta \cos \phi}$$

- (iii) Use the fact that the gradient of PS = gradient of SQ to show that **2**

$$\frac{\sin \phi - \sin \theta}{\cos \theta \sin \phi - \sin \theta \cos \phi} = \frac{1}{e}$$

- (iv) Hence or otherwise show that R lies on the directrix of the ellipse. **1**

End of Question 6

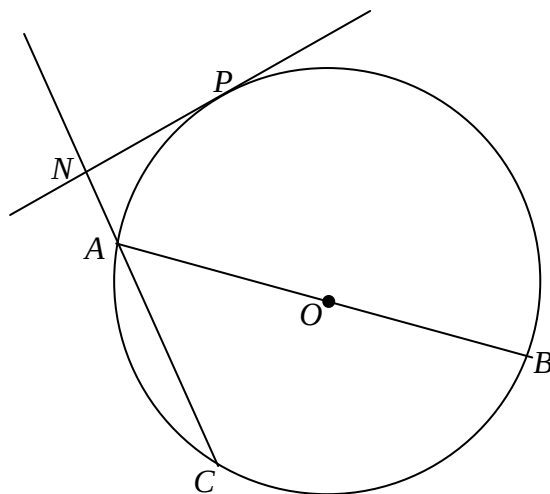
Question 7. (15 marks) Start a new page

Marks

- (a) Let α, β, γ be the roots of the equation $x^3 + qx + r = 0$. 2
Write down the cubic equation whose roots are $\alpha^{-1}, \beta^{-1}, \gamma^{-1}$.

- (b) Let ω be a non-real root of $z^7 - 1 = 0$.
- (i) Show that $1 + \omega + \omega^2 + \omega^3 + \omega^4 + \omega^5 + \omega^6 = 0$. 1
- (ii) Show that $(1 + \omega)(1 + \omega^2)(1 + \omega^4) = 1$. 1
- (iii) Simplify $(\omega + \omega^2 + \omega^4)(\omega^6 + \omega^5 + \omega^3)$ 2
- (iv) Sketch on the Argand diagram all seven roots of $z^7 - 1 = 0$ 1

- (c) In a circle centre O , a diameter AB and a chord AC are drawn.
 P is the point on the circumference on the side of AB opposite to C , such that the tangent at P is perpendicular to CA produced.
The tangent at P and the line CA produced intersect at the point N .



Copy this diagram into your examination booklet.

Prove that:

- (i) $PC = PB$ 3
- (ii) $\angle APC + 2\angle ACP = 90^\circ$ 3
- (iii) $\angle PAB = \angle NPC$ 2

End of Question 7

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Question 8. (15 marks) Start a new page

Marks

(a) (i) Sketch $y = \sec x$ in the domain $-2\pi \leq x \leq 2\pi$

1

(ii) Using a suitable domain sketch $y = \sec^{-1} x$.

2

(b) For all integers $n \geq 1$, let

$$t_n = \frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} + \dots + \frac{1}{2n-1} + \frac{1}{2n}$$

That is: $t_1 = \frac{1}{2}$

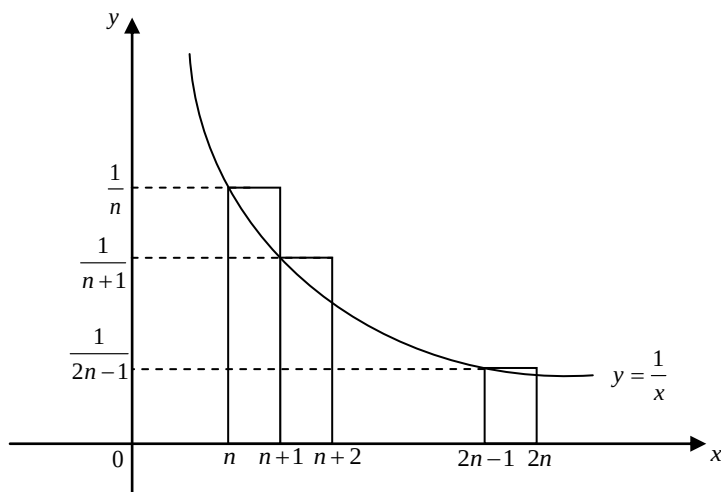
$$t_2 = \frac{1}{3} + \frac{1}{4}$$

$$t_3 = \frac{1}{4} + \frac{1}{5} + \frac{1}{6}$$

.....

(i) Show that $t_n + \frac{1}{2n} = \frac{1}{n} + \frac{1}{n+1} + \dots + \frac{1}{2n-1}$

2



The diagram above shows the graph of the function $y = \frac{1}{x}$ for $n \leq x \leq 2n$.

(ii) By using the diagram and the area of upper rectangles, show that $t_n + \frac{1}{2n} > \ln 2$

3

[Note that it can similarly be shown that $t_n < \ln 2$]

Questions 8 continued on page 13

Question 8 continued

Marks

For all integers $n \geq 1$ let

$$s_n = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{2n-1} - \frac{1}{2n}$$

That is:

$$s_1 = 1 - \frac{1}{2}$$

$$s_2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4}$$

$$s_3 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6}$$

.....

(iii) Prove by mathematical induction that $s_n = t_n$ **4**

(iv) Hence find, to three decimal places, the value of $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{9999} - \frac{1}{10000}$ **3**

End of Test

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a \neq 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2}) \quad \text{NOTE: } \ln x = \log_e x, \quad x > 0$$



KINCOPPAL–ROSE BAY
SCHOOL OF THE SACRED HEART

2009

**HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION**

Extension 2 Mathematics (Solutions)

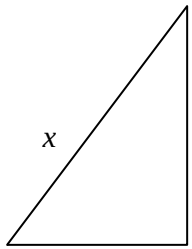
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Question	Criteria	Marks
1(a)(i)	$f(x) = x\sqrt{4-x^2}$ $f(-x) = -x\sqrt{4-(-x)^2} = -x\sqrt{4-x^2}$ $-f(x) = -x\sqrt{4-x^2}$ $\therefore f(-x) = -f(x) \text{ an odd function } \boxed{\checkmark}$	1
1(a)(ii)	$\int_{-2}^2 (x\sqrt{4-x^2} - \sqrt{4-x^2}) = \int_{-2}^2 (x\sqrt{4-x^2}) - \int_{-2}^2 (\sqrt{4-x^2})$ $= \text{odd function} - \text{semi circle} \quad \boxed{\checkmark}$ $= 0 - \frac{\pi \times 2^2}{2}$ $= 0 - 2\pi$ $= -2\pi \quad \boxed{\checkmark}$	2
(1)(b)	$\int \frac{dx}{\sqrt{4x^2+36}} = \frac{1}{2} \int \frac{dx}{\sqrt{x^2+9}} \quad \boxed{\checkmark}$ $= \frac{1}{2} \int \frac{dx}{\sqrt{x^2+9}}$ $= \frac{1}{2} \ln(x + \sqrt{x^2+9}) + c \quad \text{or} \quad \ln(2x + \sqrt{4x^2+36}) + C \quad \boxed{\checkmark}$	2
(1)(c)	Let $\frac{A}{2t+1} + \frac{B}{2-t} = \frac{5}{(2t+1)(2-t)}$ $\therefore A(2-t) + B(2t+1) = 5$ If $t = 2$, then $5B = 5 \rightarrow B = 1$ $t = -\frac{1}{2}$, then $\frac{5}{2}A = 5 \rightarrow A = 2$ $\therefore \int_0^1 \frac{5dt}{(2t+1)(2-t)} = \int_0^1 \left(\frac{2}{2t+1} + \frac{1}{2-t} \right) dt \quad \boxed{\checkmark}$ $= [\ln(2t+1) - \ln(2-t)]_0^1$ $= \left[\ln\left(\frac{2t+1}{2-t}\right) \right]_0^1 \quad \boxed{\checkmark}$ $= \ln 3 - \ln\left(\frac{1}{2}\right)$ $= \ln 6 \quad \boxed{\checkmark}$	3

(1)(d)	$t = \tan \frac{x}{2}$ $\frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2}$ $2 \cos^2 \frac{x}{2} dt = dx$ $\text{since } \cos^2 \frac{x}{2} = \frac{1}{1+t^2}$ $\therefore dx = \frac{2}{1+t^2} dt \quad \boxed{\checkmark}$	$\int \operatorname{cosec} x \, dx$ $= \int \frac{1}{\sin x} \times \frac{2}{1+t^2} dt$ $= \int \frac{1}{\frac{2t}{1+t^2}} \times \frac{2}{1+t^2} dt$ $= \int \frac{1+t^2}{2t} \times \frac{2}{1+t^2} dt$ $= \int \frac{1}{t} dt \quad \boxed{\checkmark}$ $= \ln(t) + C$ $= \ln\left(\tan \frac{x}{2}\right) + C \quad \boxed{\checkmark}$	3
(1)(e)	(e) $x = 4 \sec \theta$ $x = \frac{4}{\cos \theta}$ $\frac{dx}{d\theta} = \frac{\cos \theta \times 0 - 4 \times -\sin \theta}{\cos^2 \theta}$ $\frac{dx}{d\theta} = \frac{4 \sin \theta}{\cos^2 \theta}$ $\frac{dx}{d\theta} = 4 \tan \theta \sec \theta \quad \boxed{\checkmark}$  $\square$ $\sqrt{x^2 - 16}$ $x = 4 \sec \theta$ $\frac{x}{4} = \sec \theta$ $\frac{4}{x} = \cos \theta$ $\therefore \tan \theta = \frac{\sqrt{x^2 - 16}}{4}$	$\int \frac{\sqrt{x^2 - 16}}{x} dx$ $\int \frac{\sqrt{4^2 \sec^2 \theta - 16}}{4 \sec \theta} \times 4 \tan \theta \sec \theta d\theta$ $\int \frac{\sqrt{16(\sec^2 \theta - 1)}}{4 \sec \theta} \times 4 \tan \theta \sec \theta d\theta$ $\int 4 \tan^2 \theta \, d\theta \quad \boxed{\checkmark}$ $\int 4(\sec^2 \theta - 1) \, d\theta$ $\int 4 \sec^2 \theta - 4 \, d\theta$ $= 4 \tan \theta - 4\theta + C \quad \boxed{\checkmark}$ $= \frac{4\sqrt{16 - x^2}}{4} - 4 \cos^{-1}\left(\frac{4}{x}\right) + C$ $= \sqrt{16 - x^2} - 4 \cos^{-1}\left(\frac{4}{x}\right) + C \quad \boxed{\checkmark}$	4

Question	Criteria	Marks
2(a)	$\frac{2-5i}{4-3i} \times \frac{4+3i}{4+3i} \quad \checkmark$ $= \frac{8+6i-20i-15i^2}{16-9i^2}$ $= \frac{23-14i}{25} \quad \checkmark$	2
2(b)	$(a-ib)^2 = -21-20i$ $a^2 - 2aib + i^2b^2 = -21-20i$ $a^2 - b^2 = -21 \quad \text{and} \quad -2aib = -20i \quad \checkmark$ $\therefore a = \frac{10}{b} \Rightarrow \left(\frac{10}{b}\right)^2 - b^2 = -21$ $\frac{100}{b^2} - b^2 = -21$ $b^4 - 21b^2 - 100 = 0$ $(b^2 - 25)(b^2 + 4) = 0$ $\therefore b = \pm 5 \quad \text{and} \quad a = \frac{10}{\pm 5} = \pm 2 \quad \checkmark \checkmark$	3
2(c)	$(\sin \theta + i \cos \theta)(\cos \theta - i \sin \theta)$ $= (\sin \theta + i \cos \theta)(\cos(-\theta) + i \sin(-\theta)) \quad \checkmark$ $ z = \sqrt{\sin^2 \theta + \cos^2 \theta} = 1 \quad \checkmark$ $\arg z = -\theta \quad \checkmark$	3
2(d)(i)	$(d) (i) \left \frac{z-1}{z+1} \right = 2$ $\left \frac{x+iy-1}{x+iy+1} \right = 2$ $\frac{\sqrt{(x-1)^2 + y^2}}{\sqrt{(x+1)^2 + y^2}} = 2$ $\sqrt{(x-1)^2 + y^2} = 2\sqrt{(x+1)^2 + y^2}$ $(x-1)^2 + y^2 = 4[(x+1)^2 + y^2]$ $x^2 - 2x + 1 + y^2 = 4x^2 + 8x + 4 + 4y^2$ $3x^2 + 10x + 3y^2 + 3 = 0 \quad \checkmark$ $x^2 + \frac{10}{3}x + y^2 + 1 = 0$ $x^2 + \frac{10}{3}x + \left(\frac{5}{3}\right)^2 + y^2 = -1 + \left(\frac{5}{3}\right)^2$ $\left(x + \frac{5}{3}\right)^2 + y^2 = \frac{16}{9} \quad \checkmark$	2

2(d)(ii)

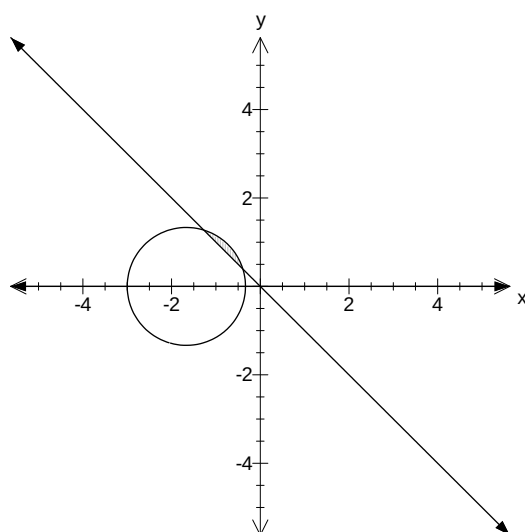
$$(ii) \left(x + \frac{5}{3}\right)^2 + y^2 \leq \frac{16}{9}$$

$$\text{centre } \left(-\frac{5}{3}, 0\right) \quad \text{radius} = \frac{4}{3}$$



$$0 \leq \arg z \leq \frac{3\pi}{4}$$

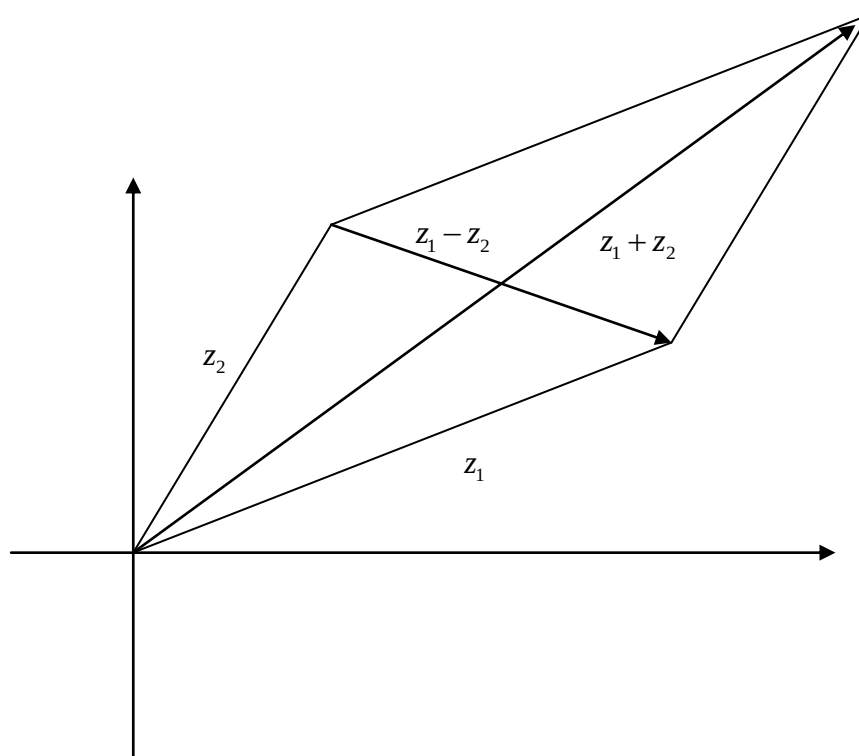
$$0 \leq y \leq -x$$



shaded region

3

2(e)

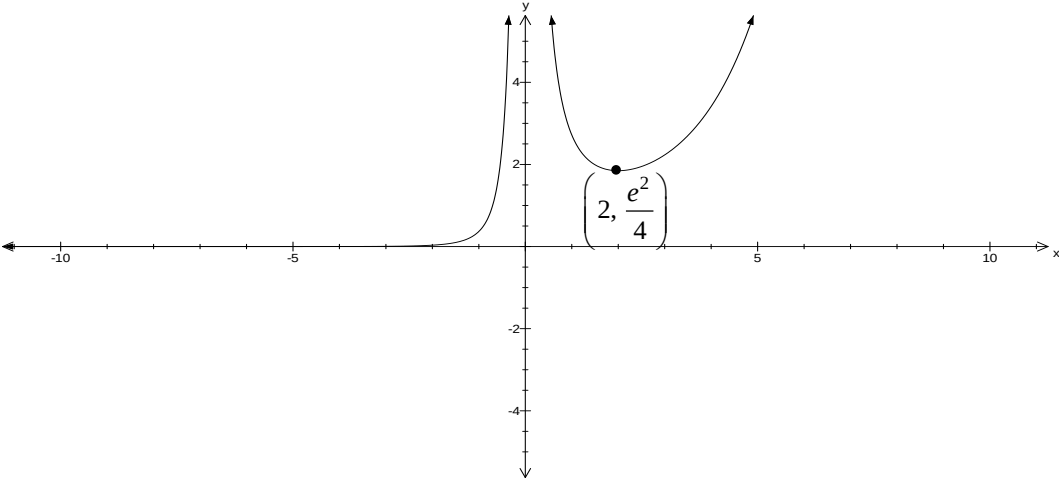
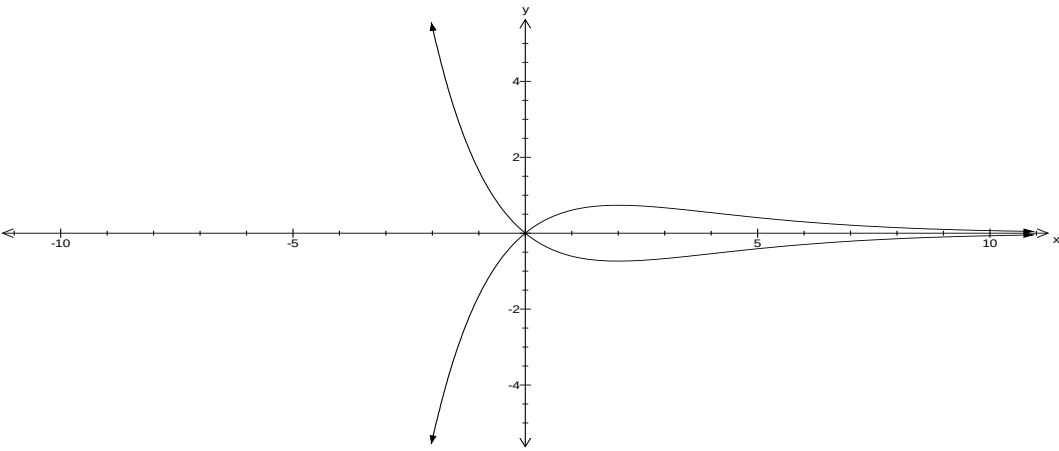
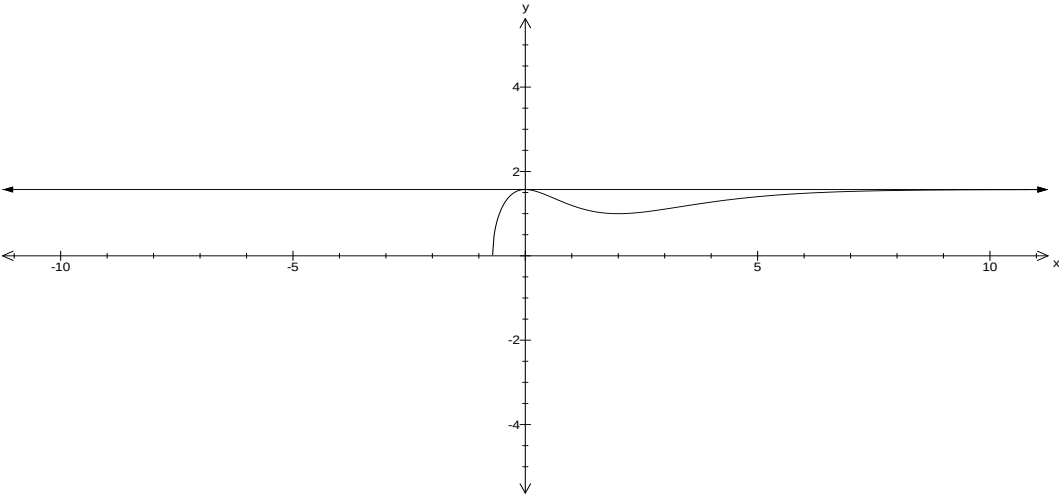


$$\text{Since } \frac{z_1 + z_2}{z_1 - z_2} = 2i \quad \text{then } \arg(z_1 + z_2) - \arg(z_1 - z_2) = \frac{\pi}{2} \quad \therefore \overrightarrow{z_1 + z_2} \perp \overrightarrow{z_1 - z_2}$$

☒ vectors z_1 , z_2 and $z_1 + z_2$
☒ vectors $z_1 - z_2$

2

Question	Criteria	Marks
3(a)(i)	☒ <i>shift up of 1 unit, asymptote to $y = 1$</i>	1
3(a)(ii)	☒ <i>reflection in y axis</i>	1
(3)(a)(iii)	☒ $x < 0$ steeper ☒ $x > 0$ graph $0 < y < 1$ (less steep gradient) and graph $y > 1$ steeper gradient	2

(3)(a)(iv)	 The graph shows a function on a Cartesian coordinate system. The x-axis ranges from -10 to 10, and the y-axis ranges from -4 to 4. There is a vertical asymptote at $x=0$. The curve approaches the asymptote from the right, goes down to a minimum turning point at $(2, \frac{e^2}{4})$, and then increases. The curve also approaches the asymptote from the left, coming from negative infinity. ☒ asymptote at $x=0$ ☒ min TP at $x=2$ and increasing gradient $x < 2$	2
3(a)(v)	 The graph shows a function on a Cartesian coordinate system. The x-axis ranges from -10 to 10, and the y-axis ranges from -4 to 4. There is a vertical asymptote at $x=0$. The curve approaches the asymptote from the right, goes down to a minimum turning point at $(2, \frac{e^2}{4})$, and then increases. The curve also approaches the asymptote from the left, coming from negative infinity. ☒ sketch only +ve section of original graph and reflect in x-axis ☒ for $-1 < y < 1$ the sketch is less steep and more steep after $y < -1$ and $y > 1$	2
(3)(a)(vi)	 The graph shows a function on a Cartesian coordinate system. The x-axis ranges from -10 to 10, and the y-axis ranges from -4 to 4. There is a vertical asymptote at $x=0$. The curve approaches the asymptote from the right, goes down to a minimum turning point at $(2, \frac{e^2}{4})$, and then increases. The curve also approaches the asymptote from the left, coming from negative infinity. ☒ range $0 \leq y \leq \frac{\pi}{2}$ ☒ asymptote to $\frac{\pi}{2}$, $x \geq 2$	2

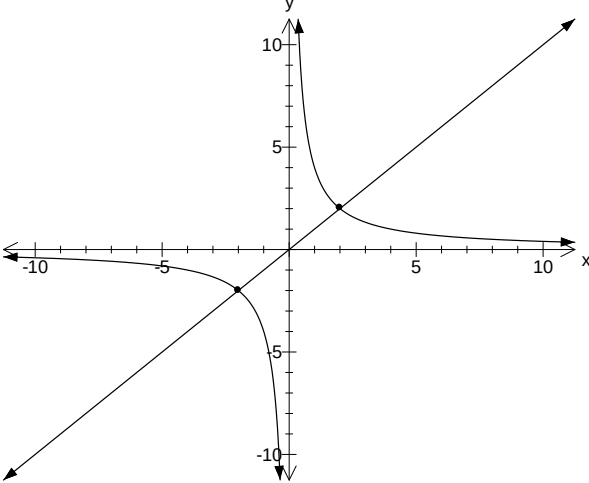
(3)(b)	$x^m y^n = k$ $u = x^m \quad v = y^n$ $u' = m x^{m-1} \quad v' = n y^{n-1} \frac{dy}{dx}$ $m y^n x^{m-1} + n x^m y^{n-1} \frac{dy}{dx} = 0 \quad \boxed{\checkmark}$ $n x^m y^{n-1} \frac{dy}{dx} = -m y^n x^{m-1}$ $\frac{dy}{dx} = \frac{-m y^n x^{m-1}}{n x^m y^{n-1}}$ $\frac{dy}{dx} = \frac{-m y^n x^m \times x^{-1}}{n x^m y^n \times y^{-1}}$ $\frac{dy}{dx} = \frac{-m y^n x^m \times y}{n x^m y^n \times x} \quad \left. \vphantom{\frac{dy}{dx}} \right\} \quad \boxed{\checkmark}$ $\frac{dy}{dx} = \frac{-m y}{n x}$	2
(3)(c)	$V = 2\pi r \times \text{thickness} \times \text{height}$ $V = 2\pi \times x \times y \times dx$ $V = 2\pi \int_0^{\sqrt{6}} x y \, dx = \quad \boxed{\checkmark}$ $V = 2\pi \int_0^{\sqrt{6}} x (6x^2 - x^4) \, dx \quad \boxed{\checkmark}$ $V = 2\pi \int_0^{\sqrt{6}} 6x^3 - x^5 \, dx$ $V = 2\pi \left[\frac{6x^4}{4} - \frac{x^6}{6} \right]_0^{\sqrt{6}}$ $V = 36\pi \quad \boxed{\checkmark}$	3

Question	Criteria	Marks
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4(a)(i)	$x^3 - 4x^2 + 2x + 5 = 0$ $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$ $= (4)^2 - 2(2)$ $= 12 \quad \boxed{\checkmark}$	1
4(a)(ii)	$x^3 - 4x^2 + 2x + 5 = 0$ $\alpha^3 - 4\alpha^2 + 2\alpha + 5 = 0$ $\beta^3 - 4\beta^2 + 2\beta + 5 = 0$ $\underline{\gamma^3 - 4\gamma^2 + 2\gamma + 5 = 0}$ $\therefore \alpha^3 + \beta^3 + \gamma^3 - 4(\alpha^2 + \beta^2 + \gamma^2) + 2(\alpha + \beta + \gamma) + 15 = 0 \quad \boxed{\checkmark}$ $\alpha^3 + \beta^3 + \gamma^3 - 4(12) + 2(4) + 15 = 0$ $\alpha^3 + \beta^3 + \gamma^3 = 48 - 8 - 15$ $= 25 \quad \boxed{\checkmark}$	2
4(b)	Monic $a=1$ and integer solutions $\therefore (x - \sqrt{2})(x + \sqrt{2})(x - a)(x - b)$ $\boxed{\checkmark}$ since sum of roots is positive then $-\sqrt{2} + \sqrt{2} + a + b > 0$ product roots = 4 $\therefore \sqrt{2} \times \sqrt{2} \times a \times b = 4$ $\left. \begin{array}{l} \boxed{\checkmark} \end{array} \right\}$ $\therefore a = -2 \quad b = 1$ Hence $P(x) = (x - \sqrt{2})(x + \sqrt{2})(x - 2)(x + 1)$ $\boxed{\checkmark}$	3
4(c)(i)	$(\cos \theta + i \sin \theta)^3 = {}^3C_0(\cos \theta)^3(i \sin \theta)^0 + {}^3C_1(\cos \theta)^2(i \sin \theta)^1 + {}^3C_2(\cos \theta)^1(i \sin \theta)^2$ $+ {}^3C_3(\cos \theta)^3(i \sin \theta)^3$ $= \cos^3 \theta + 3i \cos^2 \theta \sin \theta - 3 \cos \theta \sin^2 \theta - i \sin^3 \theta \quad \boxed{\checkmark}$ DeMoivre's theorem: $(\cos \theta + i \sin \theta)^3 = (\cos 3\theta + i \sin 3\theta) \quad \boxed{\checkmark}$ Equating real parts: $\cos 3\theta = \cos^3 \theta - 3 \cos \theta \sin^2 \theta$ $= \cos^3 \theta - 3 \cos \theta (1 - \cos^2 \theta)$ $= \cos^3 \theta - 3 \cos \theta + 3 \cos^3 \theta \quad \boxed{\checkmark}$ $= 4 \cos^3 \theta - 3 \cos \theta$	3

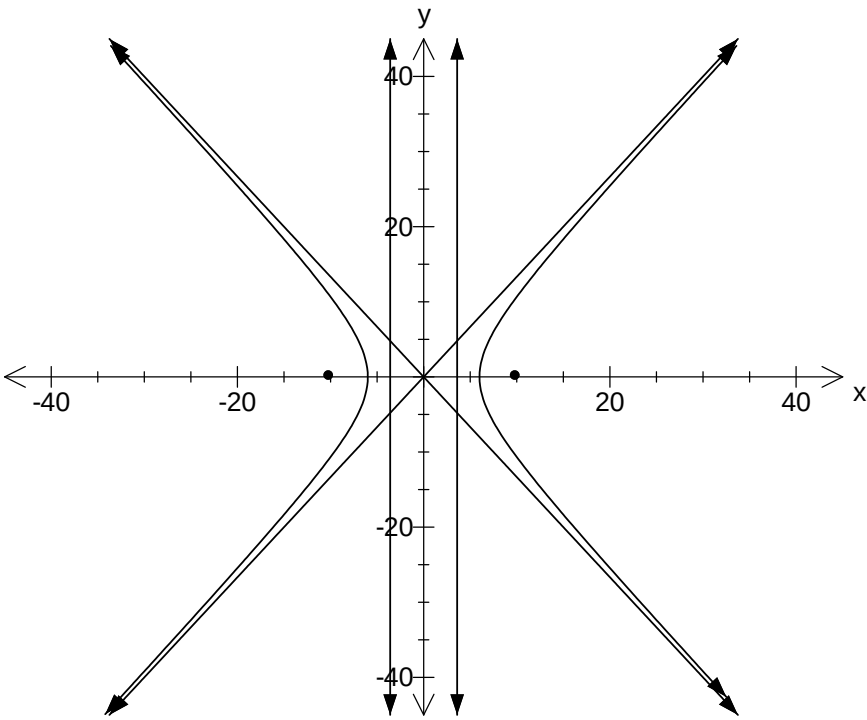
4(c)(ii)	Let $x = \cos \theta$ $8\cos^3 x - 6\cos x - 1 = 0$ $8\cos^3 x - 6\cos x = 1$ $2(4\cos^3 \theta - 3\cos \theta) = 1$ $2\cos 3\theta = 1$ $\cos 3\theta = \frac{1}{2} \quad \checkmark$ $3\theta = \cos^{-1} \frac{1}{2} \pm 2k\pi, \text{ where } k \text{ is an integer}$ $\theta = \frac{\pi}{9}(6k \pm 1) \quad \checkmark$ These values of θ give exactly three distinct values of $\cos \theta$, namely $\therefore (k=0) \quad \text{since } x = \cos \theta \Rightarrow x = \cos\left(\frac{\pi}{9}\right)$ $(k=1) \quad \text{since } x = \cos \theta \Rightarrow x = \cos\left(\frac{5\pi}{9}\right) = -\cos\left(\frac{4\pi}{9}\right)$ $(k=1) \quad \text{since } x = \cos \theta \Rightarrow x = \cos\left(\frac{7\pi}{9}\right) = -\cos\left(\frac{2\pi}{9}\right) \quad \checkmark$	
4(d)	$P(r) = 8r^3 - 4r^2 - 42r + 45 = 0$ $P'(r) = 24r^2 - 8r - 42 = 0 \quad (\text{double root}) \quad \checkmark$ $\therefore 24r^2 - 8r - 42 = 2(6r + 7)(2r - 3) = 0$ $r = \frac{-7}{6} \quad \text{or} \quad \frac{3}{2} \quad \checkmark$ $\text{sub } P\left(\frac{-7}{6}\right) = 8r^3 - 4r^2 - 42r + 45 \neq 0$ $P\left(\frac{3}{2}\right) = 8r^3 - 4r^2 - 42r + 45 = 0 \quad \text{hence } r = \frac{3}{2} \quad \checkmark$	

Question	Criteria	Marks
5(a)	$\int_1^{\infty} \frac{1}{x+1} - \frac{1}{x+3} = [\ln(x+1) - \ln(x+3)]_1^{\infty}$ $= \ln \left[\frac{x+1}{x+3} \right]_1^{\infty} \quad \boxed{\checkmark}$ $= \ln 1 - \ln \left(\frac{1}{2} \right)$ $= 0 - (\ln(1) - \ln(2))$ $= 0 - 0 + \ln 2$ $= \ln 2 \quad \left(\text{or } \ln \left(\frac{1}{2} \right) \right) \quad \boxed{\checkmark}$	2
5(b)(i)	$I_n = \int \sin^n x \, dx$ $= \int \sin x \cdot \sin^{n-1} x \, dx \quad u = \sin^{n-1} x \quad \frac{dv}{dx} = \sin x$ $\frac{du}{dx} = (n-1) \sin^{n-2} x \cos x \quad v = -\cos x$ $\therefore \int u \, dv = uv - \int v \, du$ $= -\cos x \sin^{n-1} x - \int -\cos x (n-1) \sin^{n-2} x \cdot \cos x \, dx \quad \boxed{\checkmark}$ $= -\cos x \sin^{n-1} x + (n-1) \int \cos^2 x \cdot \sin^{n-2} x \, dx$ $= -\cos x \sin^{n-1} x + (n-1) \int (1 - \sin^2 x) \sin^{n-2} x \, dx$ $= -\cos x \sin^{n-1} x + (n-1) \int \sin^{n-2} x \, dx - (n-1) \int \sin^n x \, dx \quad \boxed{\checkmark}$ $= -\cos x \sin^{n-1} x + (n-1) I_{n-2} - (n-1) I_n$ $\therefore n I_n = -\cos x \sin^{n-1} x + (n-1) I_{n-2} \quad \boxed{\checkmark}$ $\therefore I_n = -\frac{1}{n} \cos x \sin^{n-1} x + \frac{n-1}{n} I_{n-2}$	3
5b(ii)	$\int_0^{\frac{\pi}{2}} \sin^4 x \, dx = \left[-\frac{1}{4} \cos x \sin^3 x \right]_0^{\frac{\pi}{2}} + \frac{3}{4} \int_0^{\frac{\pi}{2}} \sin^2 x \, dx$ $= 0 + \frac{3}{4} \left[-\frac{1}{2} \cos x \sin x \right]_0^{\frac{\pi}{2}} + \frac{3}{4} \cdot \frac{1}{2} \int_0^{\frac{\pi}{2}} \sin^0 x \, dx \quad \boxed{\checkmark}$ $= 0 + 0 + \frac{3}{8} \int_0^{\frac{\pi}{2}} 1 \, dx$ $= \frac{3}{8} \left[x \right]_0^{\frac{\pi}{2}} = \frac{3}{8} \left[\frac{\pi}{2} - 0 \right]$ $\frac{3\pi}{16} \quad \boxed{\checkmark}$	2

5(c)(i)	 ☒ shape and where it cuts $y = x$ ie: $(2, 2)$ and $(-2, 2)$	1
5(c)(ii)	$xy = 4 \quad \therefore y = \frac{4}{x}$ $\frac{dy}{dx} = \frac{-4}{x^2}$ $\frac{dy}{dx} = \frac{-4}{4t^2} = -\frac{1}{t^2} \quad \text{at } P\left(2t, \frac{2}{t}\right) \quad \boxed{\checkmark}$ $\therefore y - \frac{2}{t} = -\frac{1}{t^2}(x - 2t) \quad \boxed{\checkmark}$ $t^2 y - 2t = -x + 2t$ $x + t^2 y = 4t$	2
5(c)(iii)	Tangent P: $x + t^2 y = 4t$ Tangent Q: $x + s^2 y = 4s$ $x = 4t - t^2 y \quad \text{and} \quad x = 4s - s^2 y$ $\therefore 4t - t^2 y = 4s - s^2 y$ $4t - 4s = t^2 y - s^2 y$ $4(t - s) = y(t^2 - s^2)$ $y = \frac{4}{t + s} \quad \boxed{\checkmark}$ $x = 4t - t^2 y$ $= 4t - t^2 \left(\frac{4}{t + s} \right)$ $= \frac{4t(t + s) - 4t^2}{t + s}$ $= \frac{4ts}{t + s} \quad \boxed{\checkmark} \quad \therefore M\left(\frac{4st}{t + s}, \frac{4}{t + s}\right)$	2

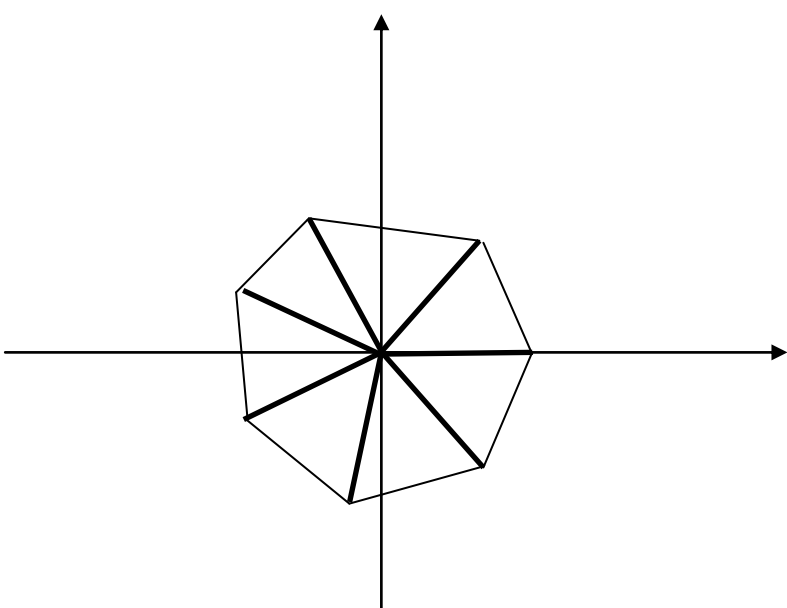
5(c)(iv)	$M\left(\frac{4st}{t+s}, \frac{4}{t+s}\right)$ $\therefore x = \frac{4st}{t+s} \text{ and } y = \frac{4}{t+s}$ $t+s = \frac{4st}{x} \text{ and } t+s = \frac{4}{y} \quad \boxed{\checkmark}$ $\text{hence } \frac{4st}{x} = \frac{4}{y} \Rightarrow y = \frac{x}{st}$ $\text{and since } s = -\frac{1}{t}$ $\therefore y = -x \text{ (straight line locus through (0,0)) } \quad \boxed{\checkmark}$ $\text{However it cannot pass through the origin as } t, s \neq 0 \quad \boxed{\checkmark}$	3
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Question	Criteria	Marks
6(a)	$y^2 = 16(1-x) \Rightarrow x = 1 - \frac{y^2}{16}$ $A = \pi(R^2 - r^2)$ $= \pi(2^2 - (2-x)^2)$ $= \pi(4x - x^2) \quad \boxed{\checkmark}$ $V = \pi \int_{-4}^4 (4x - x^2) dy \quad \boxed{\checkmark}$ $V = 2\pi \int_0^4 \left(4\left(1 - \frac{y^2}{16}\right) - \left(1 - \frac{y^2}{16}\right)^2 \right) dy$ $V = 2\pi \int_0^4 \left(4 - \frac{y^2}{4} - 1 + \frac{y^2}{8} - \frac{y^4}{256} \right) dy$ $V = 2\pi \int_0^4 \left(3 - \frac{y^2}{8} - \frac{y^4}{256} \right) dy \quad \boxed{\checkmark}$ $A = 2\pi \left[3y - \frac{y^3}{24} - \frac{y^5}{1280} \right]_0^4$ $A = 2\pi \left[\left(12 - \frac{64}{24} - \frac{1024}{1280} \right) - (0) \right]$ $A = 2\pi \left(\frac{128}{15} \right)$ $A = \frac{256\pi}{15} \quad \boxed{\checkmark}$	4
6(b)(i)	$ae = 10 \text{ and } \frac{b}{a} = \frac{4}{3}$ $\therefore a = \frac{10}{e} \text{ and } b = \frac{4a}{3} \text{ and since } b^2 = a^2(e^2 - 1)$ $\text{then } \left(\frac{4a}{3} \right)^2 = \left(\frac{10}{e} \right)^2 (e^2 - 1)$ $\left(\frac{4\left(\frac{10}{e}\right)}{3} \right)^2 = \left(\frac{10}{e} \right)^2 (e^2 - 1)$ $\frac{1600}{9e^2} = \frac{100}{e^2} (e^2 - 1)$ $\frac{16}{9} = e^2 - 1$ $e = \sqrt{\frac{16}{9} + 1} \quad \text{as } e > 1$ $e = \frac{5}{3} \quad \boxed{\checkmark}$	1

6(b)(ii)	since $e = \frac{5}{3}$ and $a = \frac{10}{e} = 6$ and $b = \frac{4a}{3} = 8$ $\therefore$ Equation of hyperbola is $\frac{x^2}{36} - \frac{y^2}{64} = 1$ ☒	1
6(b)(iii)	 ☒ shape and asymptotes $y = \pm \frac{4x}{3}$ ☒ directrices $x = \pm \frac{18}{5}$ and foci $(\pm 6, 0)$	2

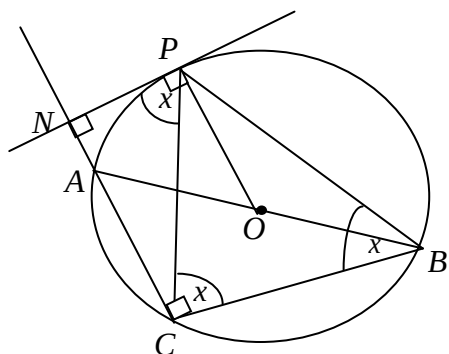
6(c)(i)	$\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0 \quad \therefore \frac{dy}{dx} = \frac{\frac{-2x}{a^2}}{\frac{2y}{b^2}} = \frac{-b^2 x}{a^2 y} = \frac{-b^2 (a \cos \theta)}{a^2 (b \sin \theta)} = -\frac{b \cos \theta}{a \sin \theta} \quad \checkmark$ Equation of Tangent : $y - b \sin \theta = \frac{b \cos \theta}{-a \sin \theta} (x - a \cos \theta)$ $-a \sin \theta y + ab \sin \theta \sin \theta = b \cos \theta x - ab \cos \theta \cos \theta$ $-a \sin \theta y - b \cos \theta x = -ab \sin \theta \sin \theta - ab \cos \theta \cos \theta$ $a \sin \theta y + b \cos \theta x = ab \sin \theta \sin \theta + ab \cos \theta \cos \theta$ $a \sin \theta y + b \cos \theta x = ab \sin^2 \theta + ab \cos^2 \theta$ $\frac{a \sin \theta y}{ab} + \frac{b \cos \theta x}{ab} = \frac{ab \sin^2 \theta}{ab} + \frac{ab \cos^2 \theta}{ab}$ $\frac{\sin \theta y}{b} + \frac{\cos \theta x}{a} = \sin^2 \theta + \cos^2 \theta \quad \checkmark$ $\frac{y \sin \theta}{b} + \frac{x \cos \theta}{a} = 1$ $\therefore$ equation of tangent at $P(a \cos \theta, b \sin \theta)$ is $\boxed{\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1}$	2
6(c)(ii)	$\therefore$ equation of tangent at $P(a \cos \theta, b \sin \theta)$ is $\boxed{\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1} \quad \dots\dots\dots \boxed{1}$ similarly the equation at $P(a \cos \phi, b \sin \phi)$ is $\boxed{\frac{x \cos \phi}{a} + \frac{y \sin \phi}{b} = 1} \quad \dots\dots\dots \boxed{2}$ $y = \frac{ab - xb \cos \theta}{a \sin \theta} \quad \text{and} \quad y = \frac{ab - xb \cos \phi}{a \sin \phi} \quad \checkmark$ $\therefore \left. \begin{aligned} \frac{ab - xb \cos \theta}{a \sin \theta} &= \frac{ab - xb \cos \phi}{a \sin \phi} \\ a^2 b \sin \phi - xab \cos \theta \sin \phi &= a^2 b \sin \theta - xab \cos \phi \sin \theta \\ a^2 b \sin \phi - a^2 b \sin \theta &= xab \cos \theta \sin \phi - xab \cos \phi \sin \theta \\ a^2 b (\sin \phi - \sin \theta) &= xab (\cos \theta \sin \phi - \cos \phi \sin \theta) \end{aligned} \right\} \checkmark$ $\therefore x = \frac{a(\sin \phi - \sin \theta)}{(\cos \theta \sin \phi - \cos \phi \sin \theta)}$	2

6(c)(iii)	$m_{PS} = \frac{b \sin \theta}{a \cos \theta - ae}$ $m_{QS} = \frac{b \sin \phi}{a \cos \phi - ae}$ since $m_{PS} = m_{QS}$ then $\frac{b \sin \theta}{a \cos \theta - ae} = \frac{b \sin \phi}{a \cos \phi - ae}$ ☒ $ab \sin \theta \cos \phi - aeb \sin \theta = ab \sin \phi \cos \theta - aeb \sin \phi$ $aeb \sin \phi - aeb \sin \theta = ab \sin \phi \cos \theta - ab \sin \theta \cos \phi$ $aeb(\sin \phi - \sin \theta) = ab(\sin \phi \cos \theta - \sin \theta \cos \phi)$ $e = \frac{\sin \phi \cos \theta - \sin \theta \cos \phi}{\sin \phi - \sin \theta}$ $\therefore \frac{\sin \phi - \sin \theta}{\sin \phi \cos \theta - \sin \theta \cos \phi} = \frac{1}{e}$ ☒	2
6(c)(iv)	Directrix of the ellipse is $x = \frac{a}{e}$ R has x-coordinates $x = \frac{a(\sin \phi - \sin \theta)}{(\cos \theta \sin \phi - \cos \phi \sin \theta)}$ $\Rightarrow \frac{x}{a} = \frac{a(\sin \phi - \sin \theta)}{(\cos \theta \sin \phi - \cos \phi \sin \theta)}$ from (iii) $\frac{\cos \phi - \cos \theta}{\sin \theta \cos \phi - \cos \theta \sin \phi} = \frac{b}{e}$ $\therefore \frac{x}{a} = \frac{b}{e}$ $x = \frac{a}{e} \quad (\text{lies on the discriminant of the ellipse})$ ☒	1

Question	Criteria	Marks
7(a)	If α, β, γ are the roots of $x^3 + qx + r = 0$ then $\alpha^{-1}, \beta^{-1}, \gamma^{-1}$ or $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$ satisfy $\left(\frac{1}{x}\right)^3 + q\left(\frac{1}{x}\right) + r = 0$ ☒ $\therefore 1 + qx^2 + rx^3 = 0$ ☒	2
7(b)(i)	ω is a root or $z^7 - 1$ hence $\omega^7 - 1 = 0$ $\omega^7 - 1 = (\omega - 1)(\omega^6 + \omega^5 + \omega^4 + \omega^3 + \omega^2 + \omega + 1) = 0$ $\omega = 1$ or $\omega^6 + \omega^5 + \omega^4 + \omega^3 + \omega^2 + \omega + 1 = 0$ ☒	1
7(b)(ii)	since $\omega = 1$ and $\omega^7 = 1$ then $(1 + \omega)(1 + \omega^2)(1 + \omega^4) = (1 + \omega^2 + \omega + \omega^3)(1 + \omega^4)$ $= 1 + \omega^4 + \omega^2 + \omega^6 + \omega + \omega^5 + \omega^3 + \omega^7$ $= 1 - 1 + 1$ ☒ $= 1$	1
7(b)(iii)	$(\omega + \omega^2 + \omega^4)(\omega^6 + \omega^5 + \omega^3) = \omega^7 + \omega^6 + \omega^4 + \omega^8 + \omega^7 + \omega^5 + \omega^{10} + \omega^9 + \omega^7$ $= 1 + \underbrace{\omega^6 + \omega^4 + \omega + 1 + \omega^5 + \omega^3 + \omega^2}_{\text{using (i)}} + 1$ ☒ $= 1 - 0 + 1$ $= 2$ ☒	2
7(b)(iv)	 ☒ $\left(\text{angle of rotation of } \frac{2\pi}{7} \text{ and } z = 1 \right)$	1

7(c)(i)

3



Let $\angle NPC = x$

$\angle NPC = \angle PBC = x$ (angles in alternate segment are equal) ☒

since $\angle ACB = 90$ (angles in a semi circle)

and $\angle CNP = 90$ ($NP \perp NC$)

$\therefore NP \parallel BC$ ☒

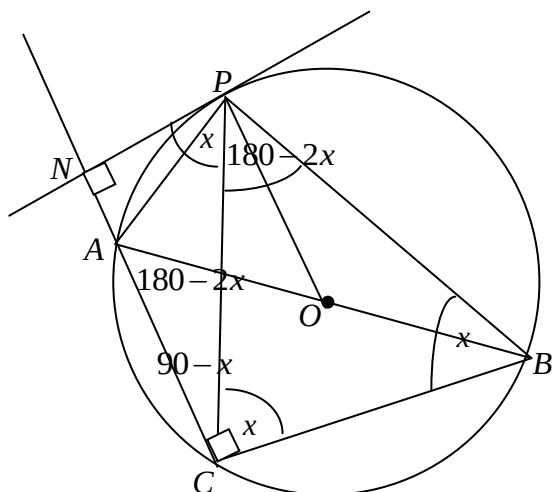
$\therefore \angle NPC = \angle PCB = x$ (alternate $\angle$'s are equal)

$\therefore \angle PCB = \angle PBC = x$ ☒

Hence $PC = PB$ (base angles of isosceles Δ are equal)

7(c)(ii)

3



$\angle NPC = \angle PCB = \angle PBC = x$

$\angle CPB = 180 - 2x$ (angle sum of a triangle is supplementary)

$\angle APB = 90$ (angles in a semi circle are right angles)

$\therefore \angle APC = 90 - (180 - 2x)$

$= 2x - 90$ (angles in a semi circle are right angles) ☒

$\angle ACP = 90 - x$ (angle in semi circle are right angles) ☒

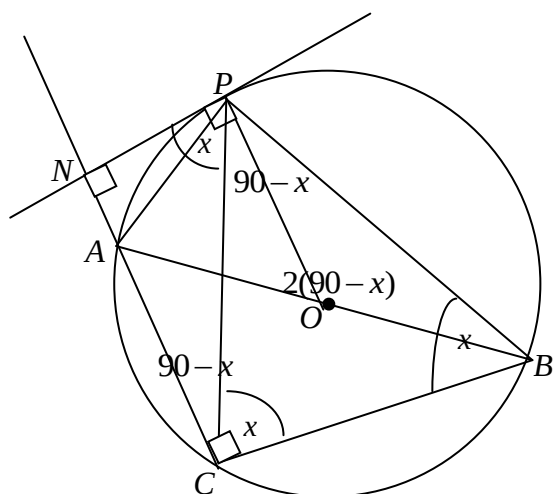
$\angle APC + 2\angle ACP = 2x - 90 + 2(90 - x)$

$= 2x - 90 + 180 - 2x$

$= 90$ ☒

7(c)(iii)

2



$$\angle NPC = \angle PCB = x \quad (\text{proven (i)})$$

$$\angle PCB = \angle PAB = x \quad (\text{angles standing on the same arc are equal})$$



$$\therefore \angle NPC = \angle PAB$$



Question	Criteria	Marks
8(a)(i)	☒ shape and asymptotes at $x = \pm \frac{\pi}{2}$ and $x = \pm \frac{3\pi}{2}$	1
8(a)(ii)	domain $0 \leq x \leq 1$ ☒ domain ☒ shape and features domain $-1 \leq x \leq 0$ ☒ domain ☒ shape and features	2

8(b)(i)	$t_n = \frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} + \dots + \frac{1}{2n-1} + \frac{1}{2n}$ $\therefore t_n + \frac{1}{2n} = \frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} + \dots + \frac{1}{2n-1} + \frac{1}{2n} + \frac{1}{2n} \quad \checkmark$ $t_n + \frac{1}{2n} = \left(\frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} + \dots + \frac{1}{2n-1} \right) + \frac{1}{n}$ $= \frac{1}{n} + \frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} + \dots + \frac{1}{2n-1} \quad \checkmark$	2
8(b)(ii)	$t_n + \frac{1}{2n} = \frac{1}{n} + \frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} + \dots + \frac{1}{2n-1}$ =sum of upper rectangles drawn on the graph ☒ $> \int_n^{2n} \frac{1}{x} dx = [\ln x]_n^{2n} \quad \checkmark$ $\therefore t_n + \frac{1}{2n} > \ln 2n - \ln n = \ln \left(\frac{2n}{n} \right) = \ln 2 \quad \checkmark$	3
8(b)(iii)	Prove true for $n=1$ $\therefore t_1 = \frac{1}{2} \quad s_1 = 1 - \frac{1}{2} = \frac{1}{2} \quad \text{Hence true for } n=1 \quad \checkmark$ Assume true for $n=k$ $\therefore t_k = s_k$ Prove true for $n=k+1$ Now $s_{k+1} = 1 - \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{2k} + \frac{1}{2(k+1)-1} + \frac{1}{2(k+1)}$ $= 1 - \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{2k} + \frac{1}{2k+1} + \frac{1}{2k+2}$ $= s_k + \frac{1}{2k+1} + \frac{1}{2k+2} \quad \checkmark$ from (i) for $n=k+1$ $t_{k+1} + \frac{1}{2(k+1)} = \frac{1}{k+1} + \frac{1}{k+2} + \frac{1}{k+3} + \frac{1}{k+4} + \dots + \frac{1}{2(k+1)-1}$ $= \frac{1}{k+1} + \frac{1}{k+2} + \frac{1}{k+3} + \frac{1}{k+4} + \dots + \frac{1}{2k+1}$ $= t_{k+1} + \frac{1}{2k+1}$ $\therefore t_{k+1} = t_k + \frac{1}{2k+1} - \frac{1}{2k+2} \quad \checkmark$ since $s_k = t_k$ and $s_{k+1} = s_k + \frac{1}{2k+1} + \frac{1}{2k+2}$ and $t_{k+1} = t_k + \frac{1}{2k+1} - \frac{1}{2k+2}$ then $s_{k+1} = t_{k+1} \quad \checkmark$ $\therefore$ By principles of mathematical induction $s_n = t_n$ for $n=1, 2, 3, \dots$	4

8(b)(iv)	$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{9999} - \frac{1}{10000} = s_{5000}$ $= t_{5000} \quad \text{by (iii)} \quad \boxed{\checkmark}$ and $\ln 2 - \frac{1}{2n} < t_n < \ln 2 \quad (\text{by (ii)}) \quad \boxed{\checkmark}$ $\therefore \ln 2 - \frac{1}{10000} < t_{5000} < \ln 2$ $\ln 2 - 0.0001 < t_{5000} < \ln 2$ $0.693147 - 0.0001 < t_{5000} < 0.693147$ $\therefore t_{5000} = 0.693 \quad (3 \text{ dec pl}) \quad \boxed{\checkmark}$	3
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